To achieve both requirements, the orbit of Mann's planet must be highly elongated. And to avoid the planet's being engulfed by Gargantua's accretion disk as it nears Gargantua, the orbit, so far as possible, must be far above or below Gargantua's equatorial plane, where the disk resides.

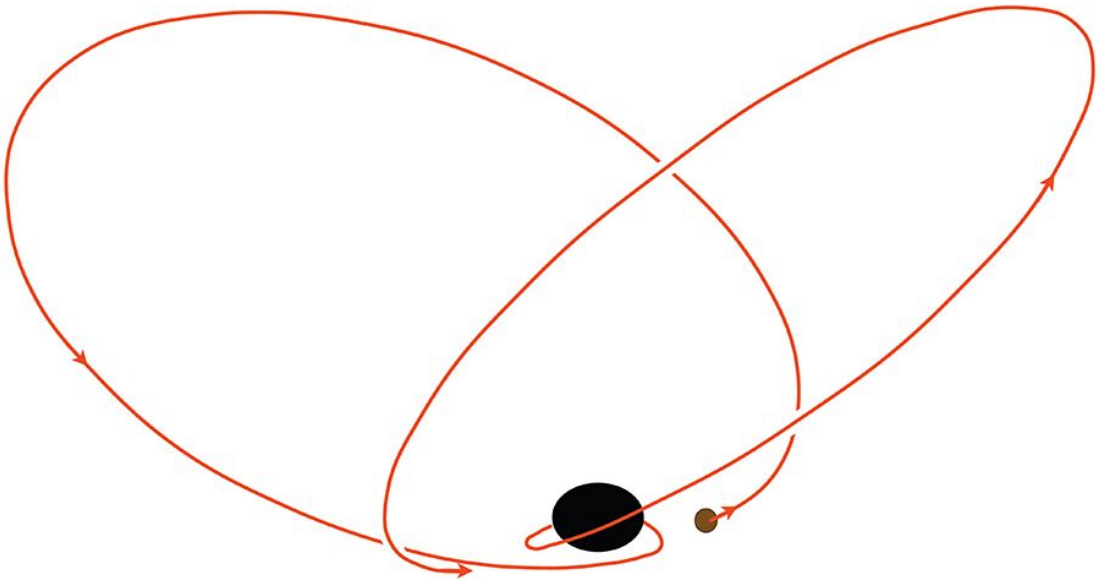


Fig. 19.1. A possible orbit for Mann's planet, computed using a highly user-friendly web application written by David Saroff; see http://demonstrations.wolfram.com/3DKerrBlackHoleOrbits.

This dictates an orbit something like that shown in Figure 19.1, though extending much farther from Gargantua, to 600 Gargantua radii or more.[33] Like the orbit of Halley's comet in our solar system (Figure 7.5), the planet swings close around Gargantua then flies out to a large distance, then returns, swings around Gargantua, and flies out again. The whirl of space near Gargantua makes the planet fly around Gargantua once or twice on each swing by, and makes its orbit precess through a large angle from one outward excursion to the next, as shown in the figure.

Mann's planet can't be accompanied by a sun on its inward and outward journeys because, when near Gargantua, huge tidal forces